

```
-- VP Planillas Database Schema --
-- === SEQUENCES === --
-- This script creates sequences for various tables in the VP Planillas database.
CREATE SEQUENCE seq_vpg_enterprise_01 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_branches_02 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_positions_03 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_employees_04 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_payroll_types_05 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_payrolls_06 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_deductions_07 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_bonuses_08 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_users_09 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_audit_logs_10 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_clock_logs_11 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;

CREATE SEQUENCE seq_vpg_labor_events_12 START
```

```
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_report_logs_13 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_report_versions_14 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_report_targets_15 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_mail_server_settings_16 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_employee_documents_17 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_payrolls_employee_18 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_employee_labor_event_19 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
CREATE SEQUENCE seq_vpg_vacations_20 START
WITH
    1 INCREMENT BY 1 NO MINVALUE NO MAXVALUE CACHE 1;
```

```
-- === END OF SEQUENCES === --
-- === TABLES === --
-- === Enterprise Structure === --
CREATE TABLE
```

```
    VPG_ENTERPRISE (
        id integer primary key,
        name varchar(50) not null,
        image bytea not null,
        creation_date date not null,
        version integer default 1 not null
    );
```

```
CREATE TABLE
```

```

VPG_BRANCHES (
    id integer primary key,
    branch_name varchar(50) not null,
    location varchar(100) not null,
    version integer default 1 not null
);

CREATE TABLE
    VPG_ENTERPRISE_BRANCH (enterprise_id INTEGER, branch_id INTEGER);

CREATE TABLE
    VPG_BRANCH_EMPLOYEE (branch_id INTEGER, employee_id INTEGER);

CREATE TABLE
    vpg_positions (
        id integer primary key default nextval ('seq_vpg_positions_03'),
        name varchar(50) not null,
        description text not null,
        base_salary decimal(10, 2) not null,
        version integer default 1 not null
    );

-- === Employee Management === --
CREATE TABLE
    vpg_employees (
        id integer primary key default nextval ('seq_vpg_employees_04'),
        first_name varchar(50) not null,
        last_name varchar(50) not null,
        middle_name varchar(50) not null,
        national_id varchar(30) not null,
        social_code varchar(100) not null,
        email varchar(100) not null,
        position_id integer not null,
        hire_date date not null,
        exit_date date,
        fired boolean default false not null,
        status char(1) not null check (status in ('A', 'I', 'T')),
        version integer default 1 not null
    );

CREATE TABLE
    vpg_employee_documents (
        id integer primary key default nextval ('seq_vpg_employee_documents_17'),
        employee_id integer not null,
        file_path varchar(255) not null,
        document_type varchar(50) not null,
        uploaded_at timestamp not null
    );

```

-- === Payroll and Deductions Management === --

CREATE TABLE

```
vpg_payroll_types (  
    id integer primary key default nextval ('seq_vpg_payroll_types_05'),  
    name varchar(50) not null,  
    description text not null,  
    version integer default 1 not null  
);
```

CREATE TABLE

```
vpg_payrolls (  
    id integer primary key default nextval ('seq_vpg_payrolls_06'),  
    payroll_type_id integer not null,  
    period_start date not null,  
    period_end date not null,  
    payment_date date not null,  
    status varchar(20) not null,  
    version integer default 1 not null  
);
```

CREATE TABLE

```
vpg_payroll_employee (  
    payroll_id integer not null default nextval ('seq_vpg_payrolls_employee_18'),  
    employee_id integer not null,  
    gross_salary decimal(10, 2) not null,  
    total_deductions decimal(10, 2) not null,  
    net_salary decimal(10, 2) not null,  
    version integer default 1 not null  
);
```

CREATE TABLE

```
vpg_deductions (  
    id integer primary key default nextval ('seq_vpg_deductions_07'),  
    name varchar(100) not null,  
    description text not null,  
    percentage decimal(5, 2),  
    fixed_amount decimal(10, 2),  
    version integer default 1 not null  
);
```

CREATE TABLE

```
vpg_vacations (  
    id INTEGER PRIMARY KEY DEFAULT nextval ('seq_vpg_vacations_20'),  
    employee_id INTEGER NOT NULL,  
    start_date DATE NOT NULL,  
    end_date DATE NOT NULL,  
    total_days INTEGER GENERATED ALWAYS AS (end_date - start_date + 1) STORED,
```

```

        paid BOOLEAN DEFAULT TRUE,
        status VARCHAR(20) DEFAULT 'Aprobado', -- Pendiente, Aprobado, Rechazado
        version INTEGER DEFAULT 1 NOT NULL
    );

```

```

CREATE TABLE
    vpg_employee_deductions (
        employee_id INTEGER NOT NULL,
        deduction_id INTEGER NOT NULL,
        payroll_id INTEGER NOT NULL, -- A cuál planilla pertenece esta deducción
        year INTEGER NOT NULL, -- Año fiscal
        month INTEGER NOT NULL, -- Mes de la deducción
        amount DECIMAL(10, 2) NOT NULL, -- Monto aplicado al empleado en esta planilla
        version INTEGER DEFAULT 1 NOT NULL,
        PRIMARY KEY (employee_id, deduction_id, payroll_id)
    );

```

```

CREATE TABLE
    vpg_bonuses (
        id INTEGER PRIMARY KEY DEFAULT nextval ('seq_vpg_bonuses_08'),
        employee_id INTEGER NOT NULL,
        payroll_id INTEGER NOT NULL, -- Planilla en la que se pagó el bono
        year INTEGER NOT NULL, -- Año contable
        month INTEGER NOT NULL, -- Mes contable
        description VARCHAR(255) NOT NULL,
        amount DECIMAL(10, 2) NOT NULL,
        granted_at DATE NOT NULL, -- Fecha en que se concedió
        version INTEGER DEFAULT 1 NOT NULL
    );

```

-- === Users and Access Control === --

```

CREATE TABLE
    vpg_users (
        id integer primary key default nextval ('seq_vpg_users_09'),
        first_name varchar(50) not null,
        last_name varchar(50) not null,
        middle_name varchar(50) not null,
        national_id varchar(30) not null,
        email varchar(100) not null,
        username varchar(50) not null,
        password varchar(255) not null,
        role varchar(20) not null,
        version integer default 1 not null
    );

```

```

CREATE TABLE
    vpg_audit_logs (
        id integer primary key default nextval ('seq_vpg_audit_logs_10'),

```

```

        user_id integer not null,
        action varchar(100) not null,
        entity varchar(100) not null,
        entity_id integer not null,
        timestamp timestamp not null,
        details text
    );

-- === Clock/Attendance Management === --
CREATE TABLE
    vpg_clock_logs (
        id integer primary key default nextval ('seq_vpg_clock_logs_11'),
        employee_id integer not null,
        timestamp timestamp not null,
        log_type varchar(10) not null,
        remarks text,
        version integer default 1 not null
    );

-- === Labor Events Management === --
CREATE TABLE
    vpg_labor_events (
        id integer primary key default nextval ('seq_vpg_labor_events_12'),
        name varchar(100) not null,
        description text not null,
        version integer default 1 not null
    );

CREATE TABLE
    vpg_employee_labor_event (
        id integer primary key default nextval ('seq_vpg_employee_labor_event_19'),
        employee_id integer not null,
        labor_event_id integer not null,
        start_date date not null,
        end_date date,
        status varchar(20) not null,
        version integer default 1 not null
    );

-- === Report Management === --
CREATE TABLE
    vpg_report_logs (
        id integer primary key default nextval ('seq_vpg_report_logs_13'),
        report_type varchar(50) not null,
        generated_by integer not null,
        generated_at timestamp not null,
        period_start date not null,
        period_end date not null,

```

```

        file_path varchar(255) not null,
        status varchar(20) not null,
        version integer default 1 not null
    );

```

CREATE TABLE

```

    vpg_report_versions (
        id integer primary key default nextval ('seq_vpg_report_versions_14'),
        report_log_id integer not null,
        created_at timestamp not null,
        file_path varchar(255) not null,
        remarks text
    );

```

CREATE TABLE

```

    vpg_report_targets (
        id integer primary key default nextval ('seq_vpg_report_targets_15'),
        institution varchar(100) not null,
        endpoint_url varchar(255) not null,
        auth_token varchar(255) not null,
        contact_email varchar(100) not null,
        version integer default 1 not null
    );

```

-- === Mail Server Settings === --

CREATE TABLE

```

    vpg_mail_server_settings (
        id integer primary key default nextval ('seq_vpg_mail_server_settings_16'),
        host varchar(100) not null,
        port integer not null,
        username varchar(100) not null,
        password varchar(255) not null,
        from_address varchar(100) not null,
        use_ssl boolean not null,
        use_tls boolean not null,
        version integer default 1 not null
    );

```

-- === END OF TABLES === --

-- === Relationships === --

```

alter table vpg_enterprise_branch add constraint
fk_vpg_enterprise_branch_enterprise_01 foreign key (enterprise_id) references
vpg_enterprise (id);

```

```

alter table vpg_enterprise_branch add constraint fk_vpg_enterprise_branch_branches_02
foreign key (branch_id) references vpg_branches (id);

```

```
alter table vpg_branch_employee add constraint fk_vpg_branch_employee_branches_03
foreign key (branch_id) references vpg_branches (id);

alter table vpg_branch_employee add constraint fk_vpg_branch_employee_employees_04
foreign key (employee_id) references vpg_employees (id);

alter table vpg_employees add constraint fk_vpg_employees_positions_05 foreign key
(position_id) references vpg_positions (id);

alter table vpg_payrolls add constraint fk_vpg_payrolls_payroll_types_06 foreign key
(payload_type_id) references vpg_payroll_types (id);

alter table vpg_payroll_employee add constraint fk_vpg_payroll_employee_payrolls_07
foreign key (payroll_id) references vpg_payrolls (id);

alter table vpg_payroll_employee add constraint fk_vpg_payroll_employee_employees_08
foreign key (employee_id) references vpg_employees (id);

alter table vpg_employee_deduction add constraint
fk_vpg_employee_deduction_employees_09 foreign key (employee_id) references
vpg_employees (id);

alter table vpg_employee_deduction add constraint
fk_vpg_employee_deduction_deductions_10 foreign key (deduction_id) references
vpg_deductions (id);

alter table vpg_employee_deductions add constraint
fk_vpg_employee_deductions_employees_01 foreign key (employee_id) references
vpg_employees (id);

alter table vpg_employee_deductions add constraint
fk_vpg_employee_deductions_deductions_01 foreign key (deduction_id) references
vpg_deductions (id);

alter table vpg_employee_deductions add constraint
fk_vpg_employee_deductions_payrolls_01 foreign key (payroll_id) references
vpg_payrolls (id);

alter table vpg_bonuses add constraint fk_vpg_bonuses_employees_01 foreign key
(employee_id) references vpg_employees (id);

alter table vpg_bonuses add constraint fk_vpg_bonuses_payrolls_01 foreign key
(payload_id) references vpg_payrolls (id);

alter table vpg_clock_logs add constraint fk_vpg_clock_logs_employees_11 foreign key
(employee_id) references vpg_employees (id);
```



```

alter table vpg_report_logs add constraint fk_vpg_report_logs_users_12 foreign key
(generated_by) references vpg_users (id);

alter table vpg_report_versions add constraint fk_vpg_report_versions_report_logs_13
foreign key (report_log_id) references vpg_report_logs (id);

alter table vpg_audit_logs add constraint fk_vpg_audit_logs_users_14 foreign key
(user_id) references vpg_users (id);

alter table vpg_bonuses add constraint fk_vpg_bonuses_employees_15 foreign key
(employee_id) references vpg_employees (id);

alter table vpg_employee_documents add constraint
fk_vpg_employee_documents_employees_16 foreign key (employee_id) references
vpg_employees (id);

alter table vpg_employee_labor_event add constraint
fk_vpg_employee_labor_event_employees_17 foreign key (employee_id) references
vpg_employees (id);

alter table vpg_employee_labor_event add constraint
fk_vpg_employee_labor_event_labor_events_18 foreign key (labor_event_id) references
vpg_labor_events (id);

alter table vpg_vacations add constraint fk_vpg_vacations_employees_01 foreign key
(employee_id) references vpg_employees (id);

-- === END OF RELATIONSHIPS === --
-- === Indexes === --
create index idx_vpg_enterprise_branch_enterprise_id on vpg_enterprise_branch
(enterprise_id);

create index idx_vpg_enterprise_branch_branch_id on vpg_enterprise_branch (branch_id);

create index idx_vpg_branch_employee_branch_id on vpg_branch_employee (branch_id);

create index idx_vpg_branch_employee_employee_id on vpg_branch_employee (employee_id);

create index idx_vpg_employees_position_id on vpg_employees (position_id);

create index idx_vpg_payrolls_payroll_type_id on vpg_payrolls (payroll_type_id);

create index idx_vpg_payroll_employee_payroll_id on vpg_payroll_employee (payroll_id);

create index idx_vpg_payroll_employee_employee_id on vpg_payroll_employee
(employee_id);

```

```
create index idx_vpg_employee_deduction_employee_id on vpg_employee_deduction
(employee_id);

create index idx_vpg_employee_deduction_deduction_id on vpg_employee_deduction
(deduction_id);

create index idx_vpg_clock_logs_employee_id on vpg_clock_logs (employee_id);

create index idx_vpg_report_logs_generated_by on vpg_report_logs (generated_by);

create index idx_vpg_report_versions_report_log_id on vpg_report_versions
(report_log_id);

create index idx_vpg_audit_logs_user_id on vpg_audit_logs (user_id);

create index idx_vpg_bonuses_employee_id on vpg_bonuses (employee_id);

create index idx_vpg_employee_documents_employee_id on vpg_employee_documents
(employee_id);

create index idx_vpg_employee_labor_event_employee_id on vpg_employee_labor_event
(employee_id);

create index idx_vpg_employee_labor_event_labor_event_id on vpg_employee_labor_event
(labor_event_id);

-- === END OF INDEXES === --
-- === End of Script === --
```